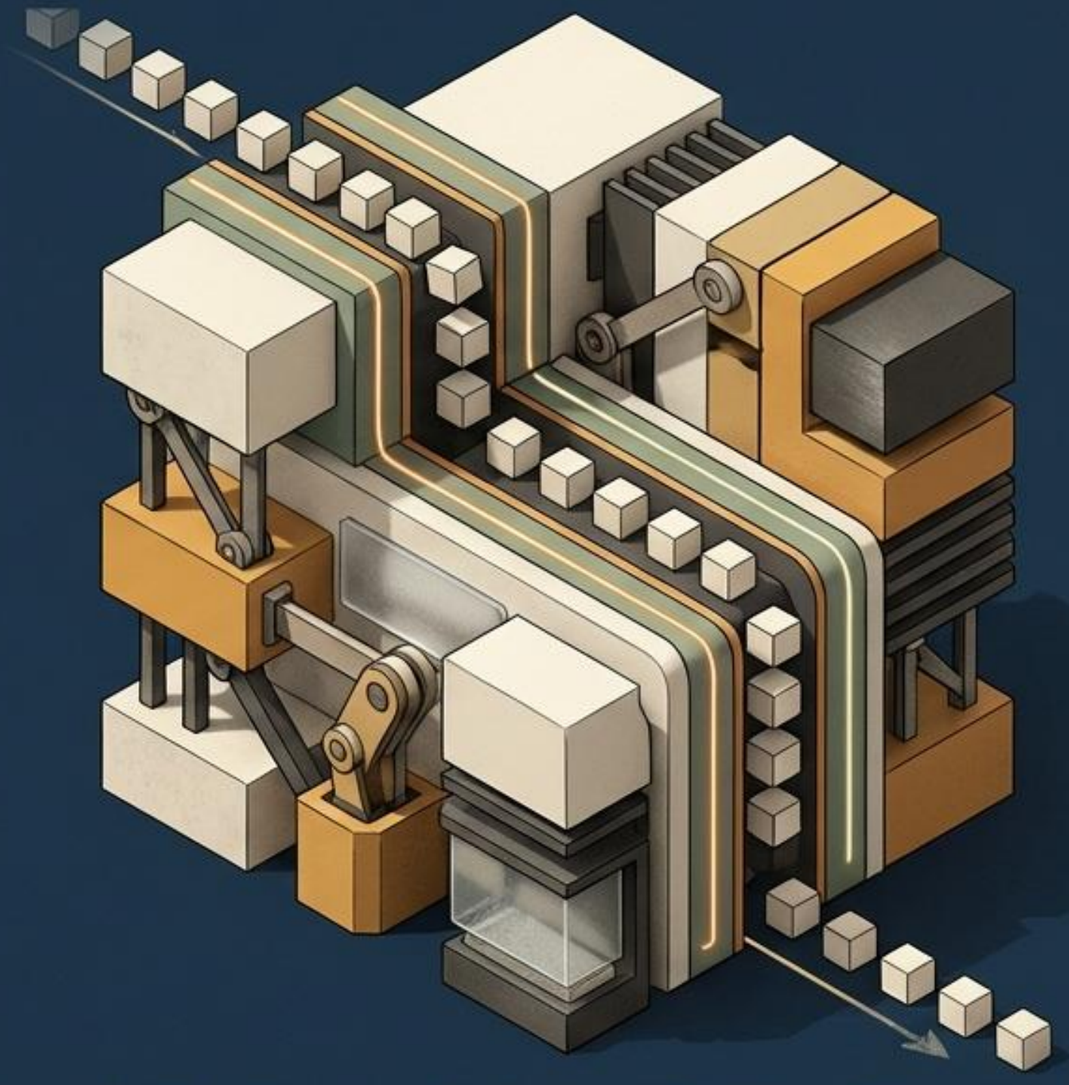
The Evolution of Intelligence

From Predictive Models to Generative AI (Day 1)

NotebookLM

AI performs tasks that mimic human behavior and decision-making



At its core, Artificial Intelligence is not about spontaneous thought; it is the mechanical replication of human problem-solving, designed to execute tasks exactly as a human would, but at an infinitely larger scale.

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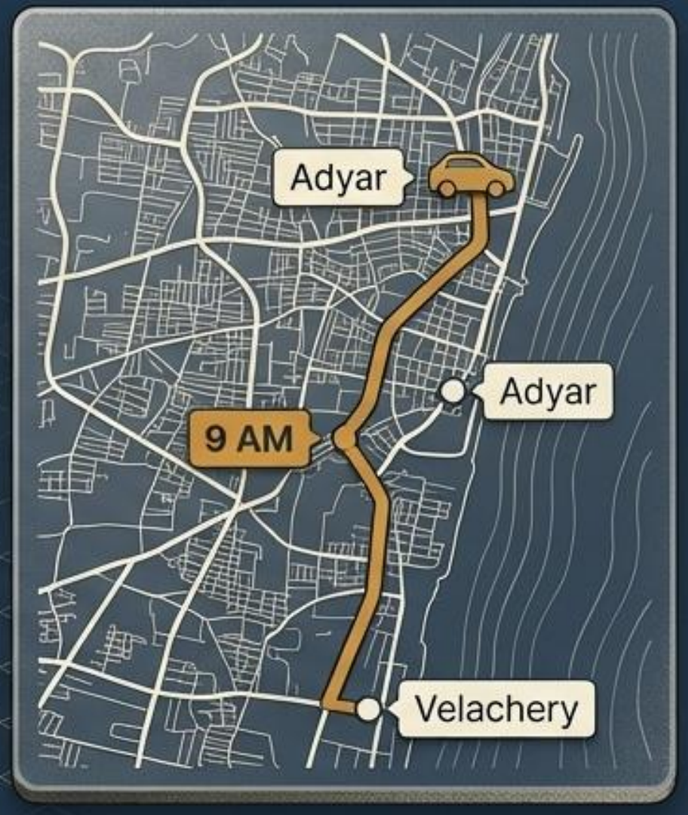
The Predictive Engine answers complex questions instantly

To accurately determine the shortest route and travel time, AI synthesizes millions of historical records alongside real-world current conditions, distilling vast complexity into one precise prediction.

Current Context:
Weather (30°C) + Traffic Density

Historical Data
Millions of past trips from 1990–2025



Accurate Prediction:
Shortest Route

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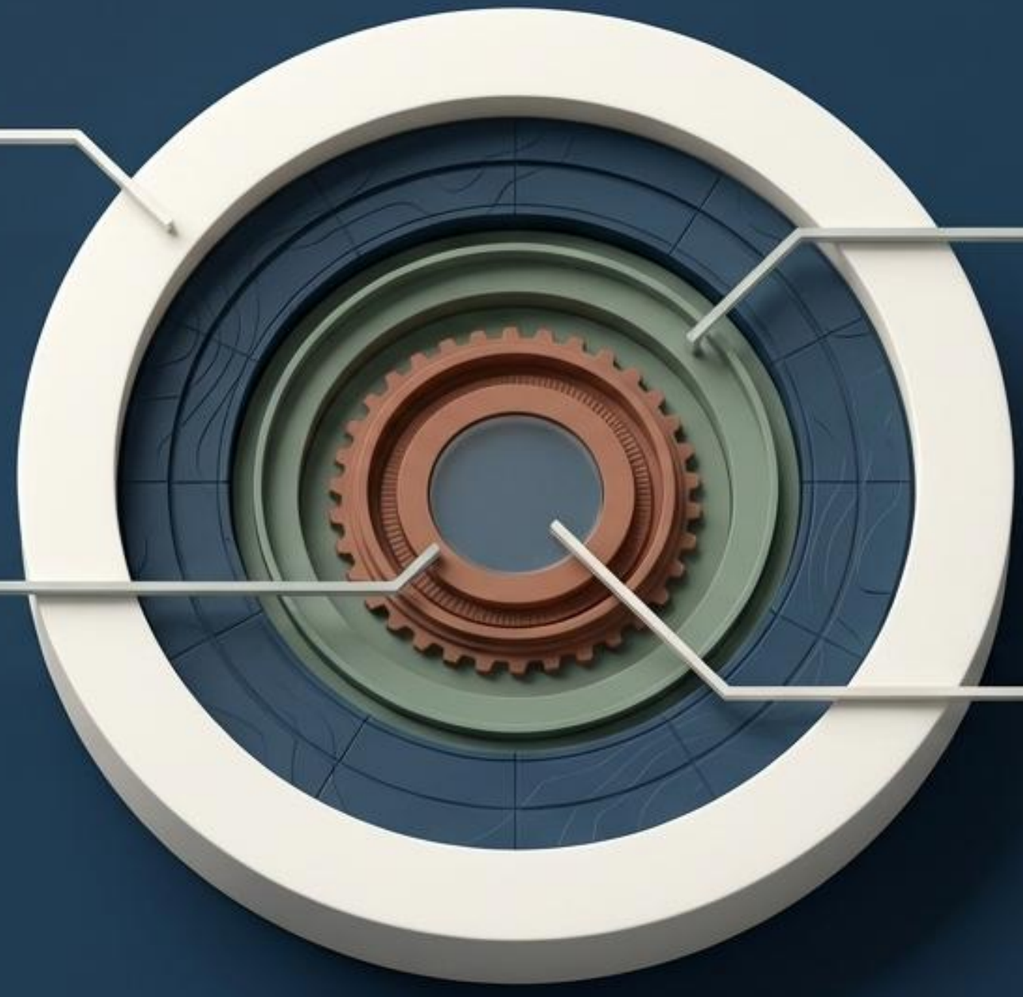
Mapping the ecosystem of artificial intelligence

Artificial Intelligence (AI)

AI: The overarching concept of machine mimicry.

Machine Learning (ML)

ML: The predictive engine driven by data.



Deep Learning (DL)

DL: The complex neural architecture.

Generative AI (Gen AI)

Gen AI: The creative spark at the core.

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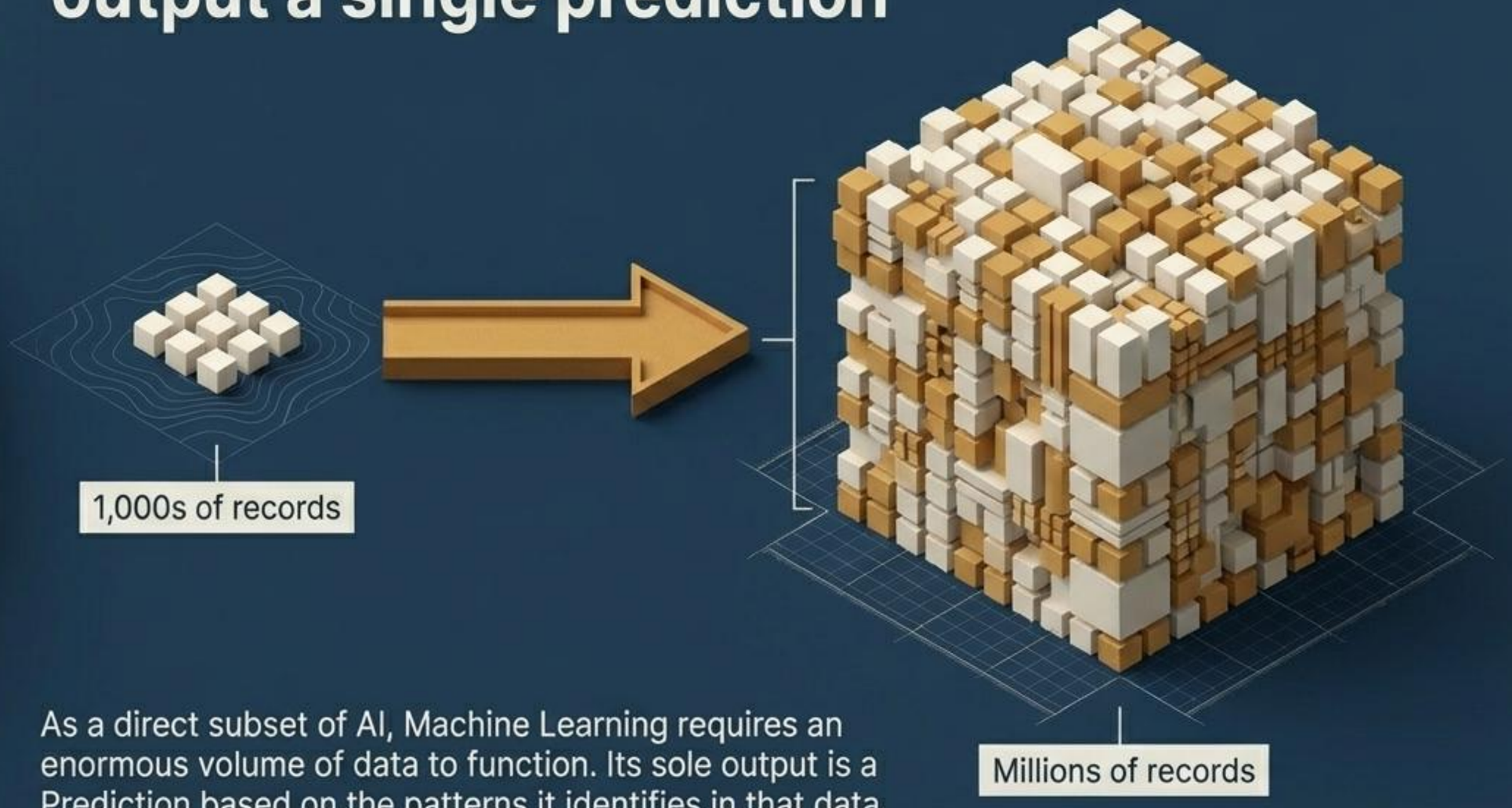
Machine Learning relies on vast historical records to output a single prediction



As a direct subset of AI, Machine Learning requires an enormous volume of data to function. Its sole output is a Prediction based on the patterns it identifies in that data.

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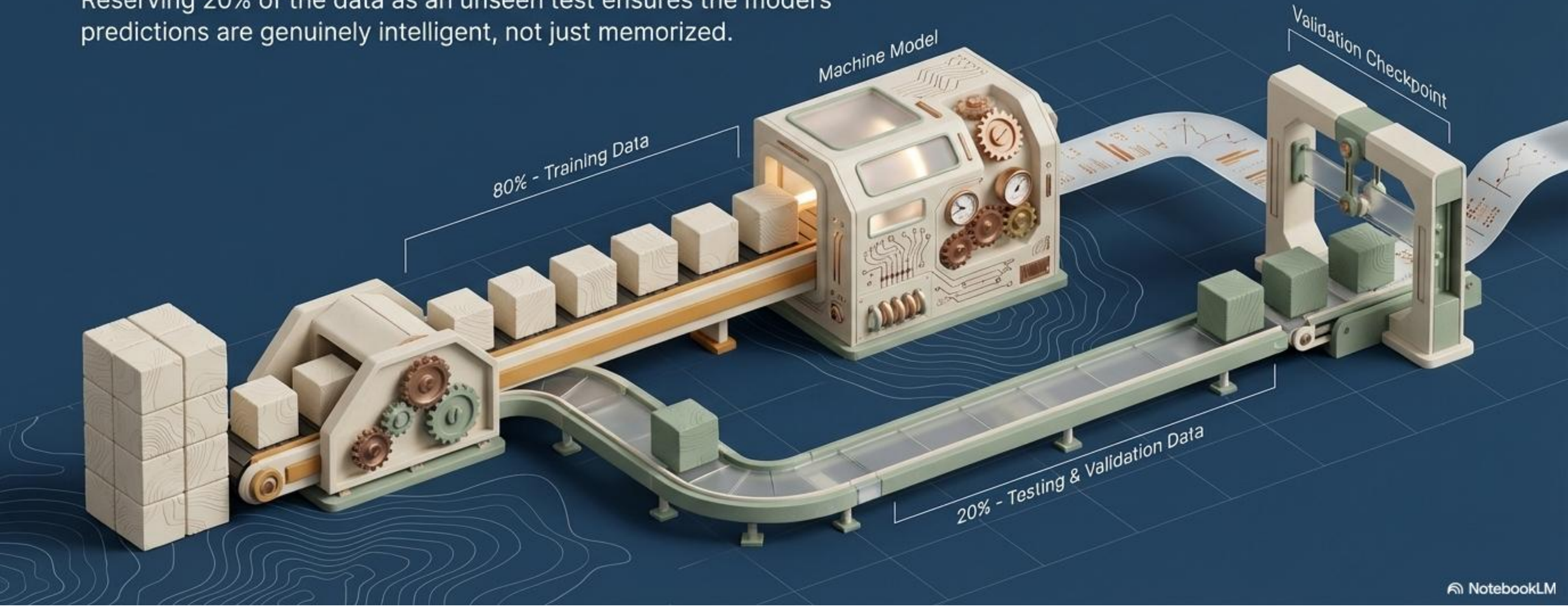
Models are trained on the majority of data and validated against unseen records

A machine cannot be accurately tested on the exact same data it used to study. Reserving 20% of the data as an unseen test ensures the model's predictions are genuinely intelligent, not just memorized.



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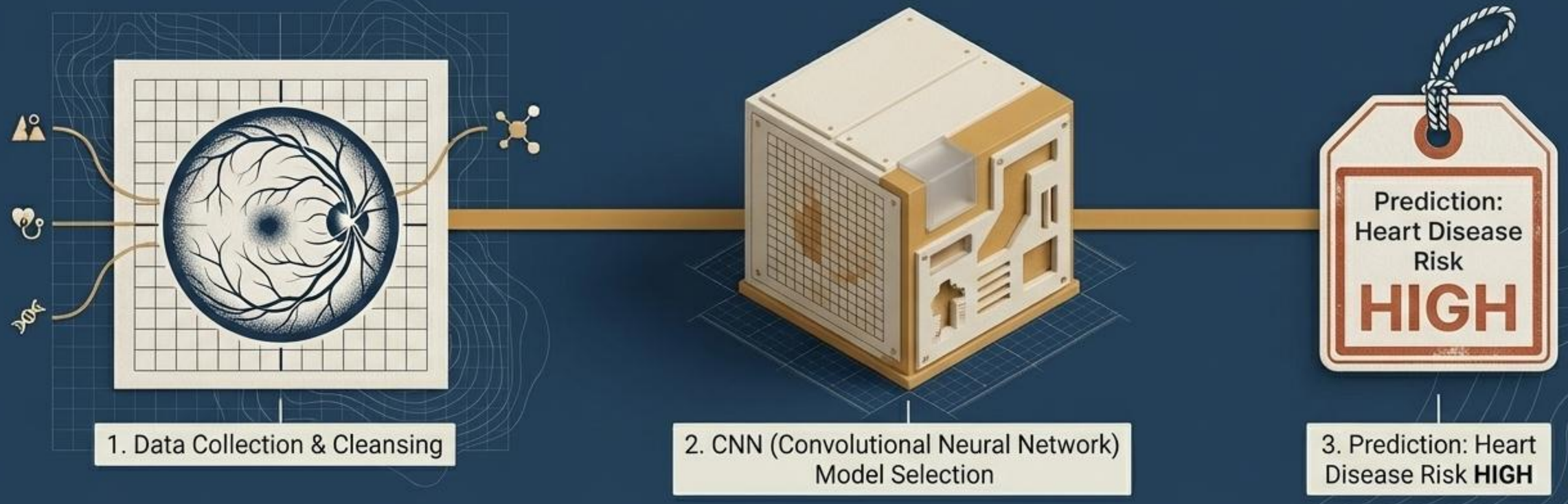
Multimodal AI ingests complex data to predict specialized health risks

Machine Learning is not limited to text or numbers. Multimodal models process millions of rows of patient parameters (age, blood pressure, genetics) alongside visual retinal imagery to output a single, life-saving predictive classification.



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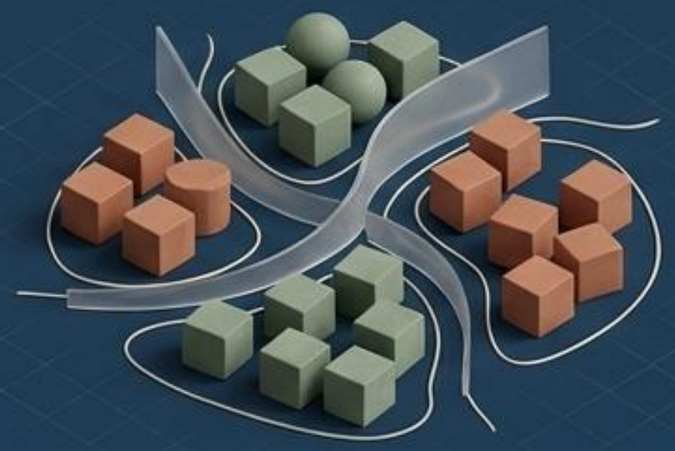
The learning method depends entirely on how the data is categorized

Supervised Learning

- Uses Labelled Data.
- Outputs are known targets.
- Content-Based (e.g., Star ratings, likes/dislikes).

Unsupervised Learning



- Uses Unlabelled Data.
- Outputs decode hidden behaviors.
- Genre/Behavior-Based (No manual tags or likes provided).

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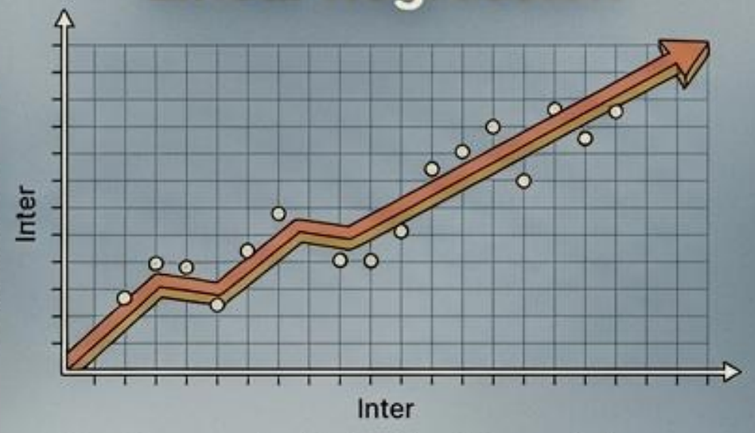
Supervised learning forecasts exact numerical values or categorizes strict risks

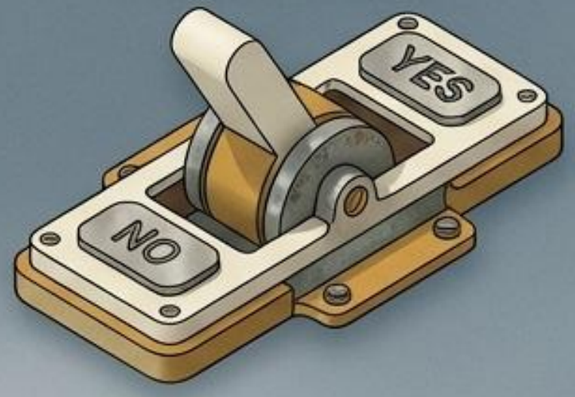
Supervised Learning

Linear Regression



Outputs numerical data. Used to predict specific, continuous values, such as real estate home purchase pricing based on historical market trends.

Classification



Outputs definitive labels or categories. Used for defect prediction testing or calculating the absolute risk of a heart attack (Yes or No).

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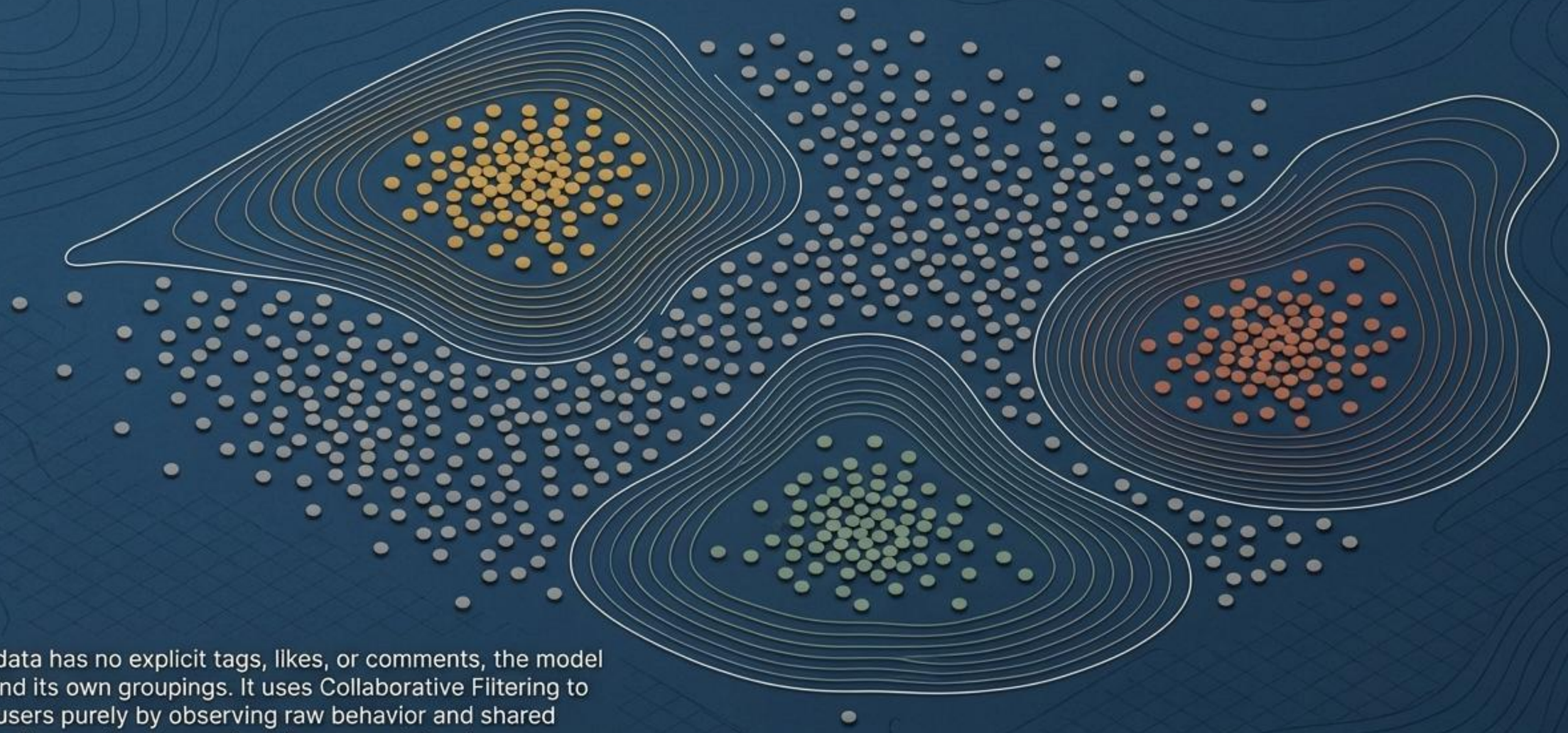
Unsupervised models decode hidden behavioral patterns without labeled instructions



When data has no explicit tags, likes, or comments, the model must find its own groupings. It uses Collaborative Filtering to group users purely by observing raw behavior and shared genres, discovering connections humans might miss.

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Deep learning integrates neural networks to process trillions of records

Deep Learning is still fundamentally a predictive engine. However, by combining vast amounts of Machine Learning data with complex Neural Networks, it can process records scaling from the millions into the trillions.

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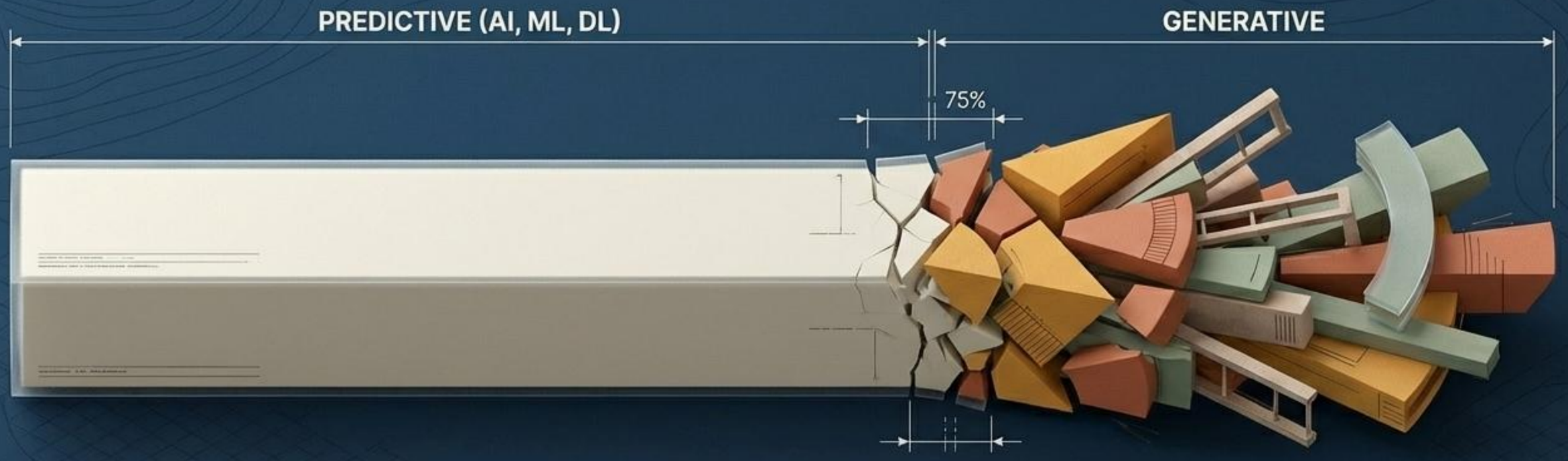
The ultimate paradigm shift moves machines from forecasting the future to creating it



Everything from routing traffic to predicting heart disease is ultimately a calculation of what already exists. Generative AI breaks this rule, stepping beyond probability to synthesize the entirely new.

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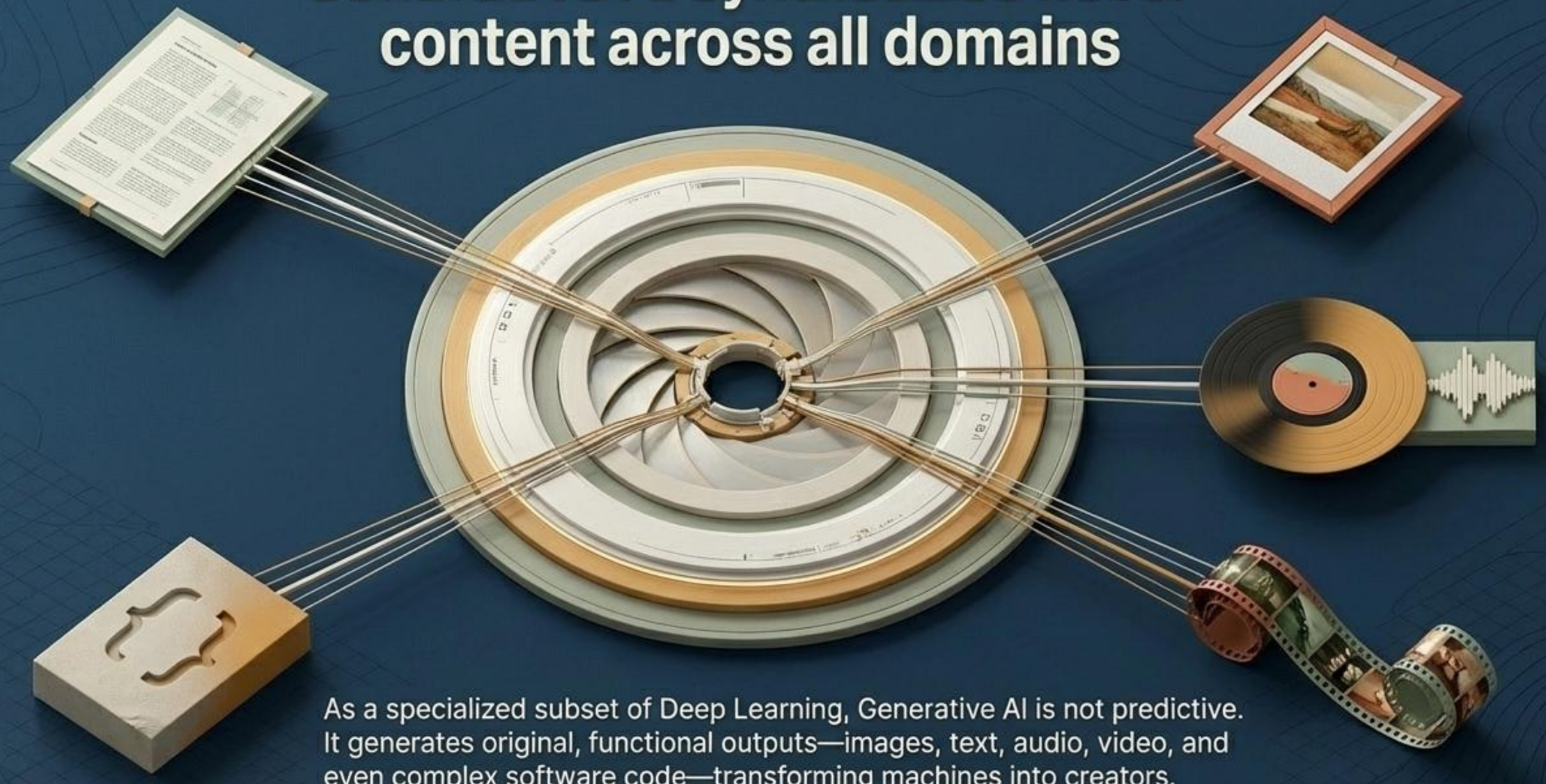
Generative AI synthesizes novel content across all domains



As a specialized subset of Deep Learning, Generative AI is not predictive. It generates original, functional outputs—images, text, audio, video, and even complex software code—transforming machines into creators.

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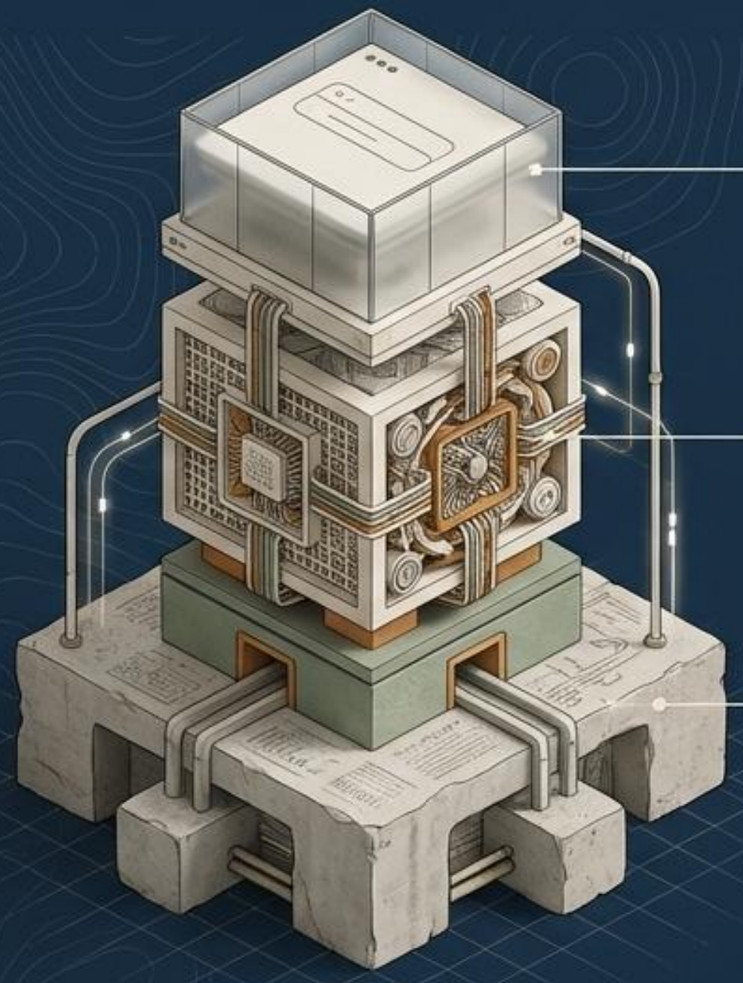
The GPT architecture scales massively to enable cross-domain generative reasoning



Top Layer:
Chat Agent (ChatGPT) by OpenAI

Middle Layer:
LLM (GPT 5.3 / 5.4)

Base Layer:
Pretrained Datasets

GPT-4

96 to 120 Layers

GPT-5

190 Layers,
120 Billion Parameters

GPT (Generative Pre-trained Transformer) models achieve unprecedented capabilities strictly through architectural scale. The jump from GPT-4 to GPT-5 demonstrates how increasing neural layers and parameters unlocks absolute generative freedom.

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The Evolution Journey: From predictive maps to generative minds

Machine & Deep Learning: Mastering Prediction	Generative AI: Mastering Creation
Vast data, strict outputs, forecasting the present	Pre-trained transformers, cross-domain synthesis, generating the future

Tomorrow — A technical deep dive into LLM anatomy and internal workings.

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